Coursecode:CSA0924

Course:Java Programming

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JAVA – THREAD SYNCHRONIZATION & COMMUNICATION

Singleton ScanEngine using Double-Checked Locking

1. AIM

To implement a thread-safe Singleton ScanEngine using Double-Checked Locking and demonstrate the importance of the volatile keyword in preventing instruction reordering under the Java Memory Model. The program also considers the safer eager initialization approach.

2. ALGORITHM

1. Start the program.

2. Declare a ScanEngine reference as a private static volatile variable.

3. Create a private constructor for the ScanEngine class to prevent direct object creation.

4. Create a public getInstance() method.

5. Check whether the instance is null without acquiring the lock.

6. If the instance is null, synchronize on the ScanEngine.class object.

7. Inside the synchronized block, check again whether the instance is null.

8. If it is still null, create a new ScanEngine object.

9. Assign the newly created object to instance.

10. Return the Singleton instance.

11. Create multiple threads that request the ScanEngine instance simultaneously.

12. Verify that all threads receive the same instance.

13. Use volatile to ensure visibility and prevent unsafe instruction reordering.

14. Consider eager initialization using class loading as an alternative.

15. Stop the program.

3. PSEUDOCODE

```
BEGIN

CLASS ScanEngine

    PRIVATE constructor

    STATIC VOLATILE instance = NULL
```

```
    METHOD getInstance()

        IF instance IS NULL THEN

            SYNCHRONIZE on ScanEngine.class

                IF instance IS NULL THEN
                    instance = CREATE NEW ScanEngine
                END IF

            END SYNCHRONIZE

        END IF

        RETURN instance

    END METHOD

END CLASS


CLASS Main

    METHOD main()

        CREATE multiple threads

        FOR each thread
            CALL ScanEngine.getInstance()
            DISPLAY instance reference
        END FOR

        WAIT for all threads to complete

        IF all references are the same THEN
            DISPLAY "Singleton instance verified"
        ELSE
            DISPLAY "Multiple instances detected"
        END IF

    END METHOD

END CLASS

END
```

4. JAVA PROGRAM

```
class ScanEngine {

    // volatile prevents instruction reordering
    // and guarantees visibility between threads
    private static volatile ScanEngine instance;
```

```
    // Private constructor
    private ScanEngine() {
        System.out.println("ScanEngine constructor executed");

        // Simulating initialization of the signature database
        try {
            Thread.sleep(100);
        } catch (InterruptedException e) {
            Thread.currentThread().interrupt();
        }

        System.out.println("Signature database initialization completed");
    }

    // Double-Checked Locking
    public static ScanEngine getInstance() {

        if (instance == null) {

            synchronized (ScanEngine.class) {

                if (instance == null) {
                    instance = new ScanEngine();
                }
            }
        }

        return instance;
    }
}


public class Main {

    public static void main(String[] args) {

        Thread t1 = new Thread(() -> {
            ScanEngine engine = ScanEngine.getInstance();
            System.out.println(
                "Thread 1 received: " +
                System.identityHashCode(engine)
            );
        });

        Thread t2 = new Thread(() -> {
            ScanEngine engine = ScanEngine.getInstance();
            System.out.println(
                "Thread 2 received: " +
                System.identityHashCode(engine)
            );
        });

        Thread t3 = new Thread(() -> {
            ScanEngine engine = ScanEngine.getInstance();
```

```
            System.out.println(
                "Thread 3 received: " +
                System.identityHashCode(engine)
            );
        });

        Thread t4 = new Thread(() -> {
            ScanEngine engine = ScanEngine.getInstance();
            System.out.println(
                "Thread 4 received: " +
                System.identityHashCode(engine)
            );
        });

        // Start all threads
        t1.start();
        t2.start();
        t3.start();
        t4.start();

        try {
            t1.join();
            t2.join();
            t3.join();
            t4.join();
        } catch (InterruptedException e) {
            Thread.currentThread().interrupt();
        }

        // Verify singleton
        ScanEngine a = ScanEngine.getInstance();
        ScanEngine b = ScanEngine.getInstance();

        if (a == b) {
            System.out.println("Singleton instance verified successfully.");
        } else {
            System.out.println("Multiple instances detected.");
        }
    }
}
```

5. IMPLEMENTATION

The program is implemented in Java using the Double-Checked Locking Singleton pattern. The ScanEngine constructor is private, allowing only one instance of the class to be created. The getInstance() method first checks whether the instance exists without synchronization. If the instance is not available, synchronization is applied and the condition is checked again before creating the object.

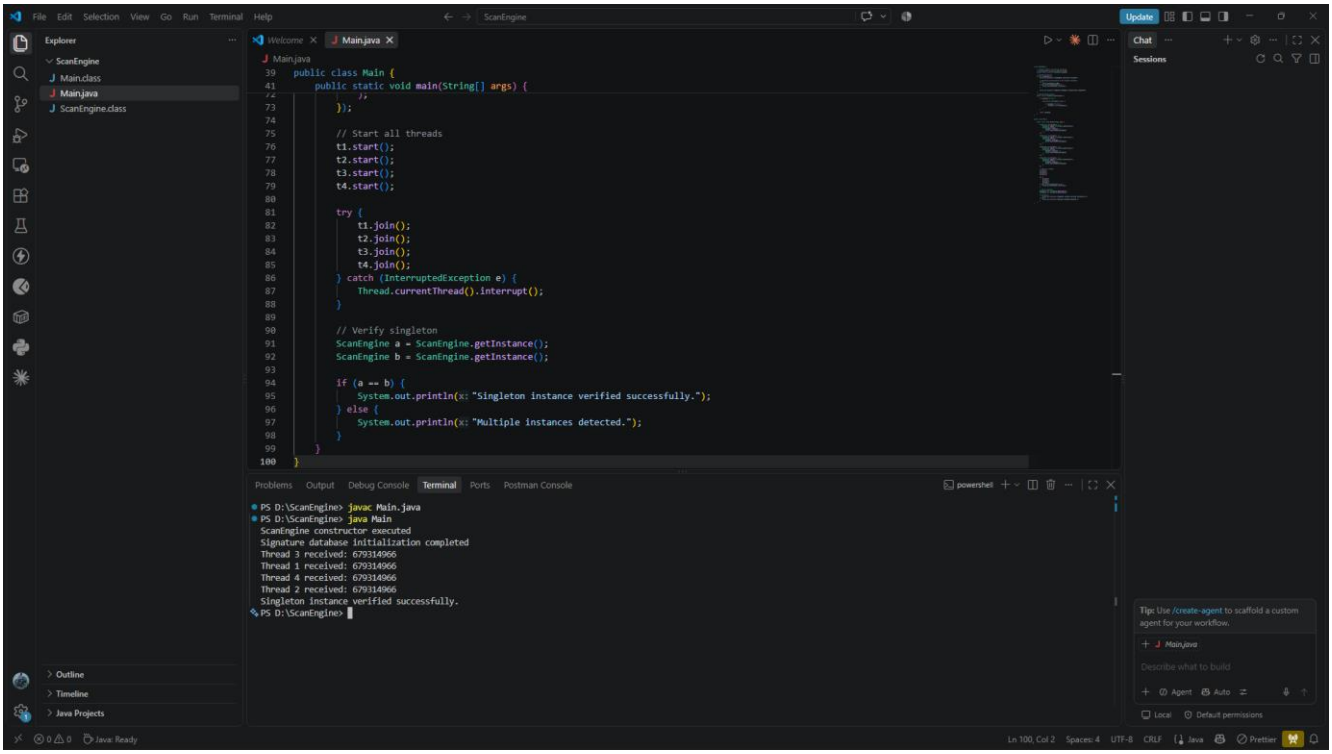
The instance variable is declared using the volatile keyword. This is essential because object creation is not necessarily a single atomic operation. It can conceptually involve memory allocation, object initialization, and assignment of the reference. Without volatile, the Java Memory Model can permit

unsafe reordering, allowing another thread to observe a non-null reference before initialization has completely finished.

The volatile keyword provides the required visibility and ordering guarantees, preventing unsafe publication of a partially initialized ScanEngine object.

6. OUTPUT

The output is to be obtained by executing the program in VS Code. The identity hash-code value may differ between executions.



7. RESULT

Thus, the thread-safe Singleton ScanEngine was successfully implemented using Double-Checked Locking. The use of the volatile keyword prevents unsafe instruction reordering and ensures that all threads observe a completely initialized ScanEngine object. The program also verifies that only a single instance is created and shared among multiple threads.